Critical inverse moments and local observation geometry on finite flat lattices

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Abstract

This appendix consolidates one sufficient coefficient condition and its limitations in the existing flat-lattice observation-rank dossier. A global inverse-conductance moment of order $D/2$ suffices for the mean-budget rank order through an actual real spectral observation. Every lower positive inverse exponent admits explicit divergence-free counterexamples. At the critical exponent, spatial observations by means on a full dyadic cube tree can require logarithmically many blocks even for a fixed weak region; the same profiles admit a constant-size unrestricted rectangular partition. The finite Fourier and flow adapters are proved analytically, and exact finite diagnostics are recorded separately from formal verification. The spectral mechanism is classical; independent novelty, an interacting gauge-theory gap and a continuum limit are not established. This is a consolidation appendix, not a separate paper or AIRR deposit.

1 Actual objects and the original observation problem

Fix $D \geq 3$, $N_c \geq 2$, $c > 0$, and the unit-spacing torus $X_n = (\mathbb{Z}/n\mathbb{Z})^D$. The actual real adjoint coefficient space has dimension $g = N_c^2 - 1$; there are D field components, so write $h = Dg$. For the actual forward difference $d_i u(x) = u(x + e_i) - u(x)$, take one scalar total coefficient $a(x) = 1 + w_{i,x,j} > 0$ shared by all derivative directions and field components. Use counting norms and the actual componentwise flat energy

$$Q_w(U) = \mathcal{E}_a(U) = \sum_{x,i,j} a(x) \|d_i U_j(x)\|^2. \tag{1}$$

The existing flat Hodge identity identifies these forms; its formal source is listed in Section 5. All new adapters in this appendix remain analytic. There is no substitution of a covariant or interacting derivative in (1).

The original local data consist of a differentiable map C at zero, with $C(0) = 0$, into an actual finite-dimensional real normed space F, and a penalty satisfying $\Phi(y) \leq K\|y\|^2$ near zero, such that

$$c\|U\|^2 \leq Q_w(U) + \Phi(C(U)) \tag{2}$$

for sufficiently small actual divergence-zero fields. "Local" here means a configuration-space neighbourhood; spatial support of C is an additional condition. The constant c is fixed, while observation maps and neighbourhoods may depend on n.

A useful necessary condition follows directly. If an actual J-dimensional divergence-zero subspace has $Q_w(U) \leq \lambda\|U\|^2$ with $\lambda < c$, then $\dim F \geq J$. Otherwise a nonzero U in that subspace lies in $\ker DC(0)$. Along tU, $C(tU) = o(t)$ and the penalty is at most $o(t^2)$. Dividing (2) by t^2 and taking $t \to 0$ contradicts $\lambda < c$. No rank or band certificate is supplied as an extra hypothesis in this argument.

2 The critical global moment theorem

Put $p = D/2 > 1$. For $m \geq 1$ suppose that

$$n^{-D} \sum_{x \in X_n} (m/a(x))^p \leq M < \infty. \tag{3}$$

Theorem 1. *Under* (3)*, there is an actual real linear spectral observation C such that*

$$c\|U\|^2 \leq Q_w(U) + c\|C(U)\|^2 \quad \textit{for every actual cochain } U,$$

and

$$\dim F \leq C_* \max\{1, (n/\sqrt{1+m})^D\}, \tag{4}$$

where, with $I_p = (p-1)^2/[p(p+1)]$ and $\Gamma_p = \frac{p}{p-1} I_p^{-(p-1)}$, one safe choice is

$$C_* = h\left[1 + \Gamma_p (3/4)^D D^{p+1} c^p M 2^p\right].$$

Thus $\Phi(y) = c\|y\|^2$, $K = c$ and any positive neighbourhood radii supply the original local data. In particular the assertion holds on the actual divergence-zero subspace.

The original subclass $1 \leq a \leq 2m$ with exact arithmetic mean m is included, but those extra normalizations are not needed for sufficiency. The distribution condition (3) remains additional; it is not derived from an arithmetic mean.

2.1 A finite normalized Fourier lemma

For a finite abelian group of size V, let $\mathcal{F}$ be its counting-unitary Fourier matrix, whose entries have modulus $V^{-1/2}$. For a frequency multiplier g put

$$L = V^{-1} \sup_{t>0} t^p \#\{k : |g(k)|^2 > t\}.$$

For a nonnegative matrix B, write $\mathsf{n}(s, B)$ for its number of eigenvalues strictly greater than s, counted with ordinary multiplicity.

Lemma 1. *For $K = M_f \mathcal{F}^* M_g \mathcal{F}$ and $s > 0$,*

$$\mathsf{n}(s, K^*K) \leq \Gamma_p L s^{-p} \sum_x |f(x)|^{2p}.$$

The trace and eigenvalue counts are not volume-normalized.

Proof. Remove phases by unitaries and set $A = \mathcal{F}^* M_{|g|^2} \mathcal{F}$. If $L = 0$ the assertion is immediate. Otherwise let $B_0 = pL/(p-1)$ and let A^+ be the inverse on $\operatorname{ran} A$, zero on its kernel. For $0 \leq \gamma \leq A$ write $\rho(x) = \langle \delta_x, \gamma \delta_x \rangle$. Positivity makes γ annihilate $\ker A$. On the range let $P_E = \mathbf{1}_{\{A^+ \leq E\}}$ and $R_E = \mathbf{1}_{\{A^+ > E\}}$. The weak-frequency tail gives

$$\langle \delta_x, P_E A P_E \delta_x \rangle = V^{-1} \sum_{|g(k)|^2 \geq 1/E} |g(k)|^2 \leq B_0 E^{p-1}.$$

Indeed the tail above t is t times its count plus the integral of counts above t; the resulting bound is $VLp\, t^{1-p}/(p-1)$. Non-strict thresholds follow by limits. The triangle inequality for $\gamma^{1/2}\delta_x$ split by $P_E + R_E$ yields

$$\langle \delta_x, R_E \gamma R_E \delta_x \rangle \geq \left(\sqrt{\rho(x)} - \sqrt{B_0} E^{(p-1)/2}\right)_+^2.$$

Exact spectral layer integration consequently gives

$$\operatorname{Tr}(A^{+}\gamma) \geq I_p B_0^{-1/(p-1)} \sum_x \rho(x)^{p/(p-1)}. \tag{5}$$

The scalar integral reduces to $\int_0^1 (1-t^{(p-1)/2})^2 dt = I_p$. Now take P to be the spectral projection of $A^{1/2} M_{|f|^2} A^{1/2}$ above s, with rank $\nu > 0$, and set $\gamma = A^{1/2} P A^{1/2}$. Then $0 \leq \gamma \leq A$, $\operatorname{Tr}(A^{+}\gamma) \leq \nu$, and

$$s\nu < \sum_x |f(x)|^2 \rho(x) \leq \Big(\sum_x |f(x)|^{2p}\Big)^{1/p} \Big(\sum_x \rho(x)^{p/(p-1)}\Big)^{(p-1)/p}.$$

Insert (5) and raise to p. The constant becomes $B_0 I_p^{-(p-1)} = \Gamma_p L$. This also explicitly handles a zero-frequency kernel. □

The density method is classical and follows the Rumin–Frank approach [1, Section 2]. The proof above supplies its finite normalization and kernel adapter; it is not a claim of a new Cwikel mechanism or independent discovery.

2.2 Actual lattice frequencies and the weighted inverse

For $n \geq 2$ let $L_0 = d^* d$ and work first on the actual counting-mean-zero scalar space H_0. Its Fourier symbols are

$$z_i(k) = e^{2\pi i k_i/n} - 1, \qquad \lambda(k) = 4 \sum_i \sin^2(\pi k_i/n).$$

Choose centred integer representatives r_i with $|r_i| \leq n/2$, using only one representative at an even endpoint. Concavity of sine implies

$$\lambda(k) \geq 16|r|^2/n^2, \qquad |g_i(k)| := |z_i(k)/\lambda(k)| \leq n/(4|r|), \quad k \neq 0.$$

Set $g_i(0) = 0$. If $|g_i| > \tau$, then $0 < |r| < R = n/(4\tau)$. For $R \leq 1$ there are none, and for $R > 1$ there are at most $(3R)^D$ representatives. Thus

$$n^{-D} \sup_{t>0} t^{D/2} \#\{|g_i|^2 > t\} \leq (3/4)^D. \tag{6}$$

This is uniform in n, including $n = 2$, not an imported infinite-volume bound.

Let $L_a = d^* M_a d$. On H_0 it is positive definite. For $f \in H_0$, the flow $j_* = M_a d L_a^{-1} f$ minimizes $\langle j, M_{a^{-1}} j\rangle$ among $d^* j = f$: write $j = j_* + k$, $d^* k = 0$, and the cross term is $\langle dL_a^{-1} f, k\rangle = 0$. Inserting the explicit feasible flow $j = dL_0^{-1} f$ gives the actual quadratic-form comparison

$$L_a^{-1} \leq \sum_{i=1}^{D} T_i^* T_i, \qquad T_i = M_{a^{-1/2}} d_i L_0^{-1}. \tag{7}$$

Extend L_0^{-1} by zero on constants. Lemma 1 and (6) bound each $\mathsf{n}(s, T_i^* T_i)$ by

$$\Gamma_p (3/4)^D s^{-p} \sum_x a(x)^{-p}.$$

For nonnegative matrices, the count of their sum above t is no larger than the sum of their counts above t/D: intersect the complements of their high spectral subspaces and apply min–max.

No commutation is required. With $t = 1/c$ in (7), inversion on H_0 and the single scalar constant mode yield

$$\#\{\lambda(L_a) < c\} \le 1 + \Gamma_p(3/4)^D D^{p+1} c^p M n^D / m^p. \tag{8}$$

No spectral-band dimension is assumed in this derivation.

Completion of Theorem 1. Choose an actual real counting-orthonormal eigenbasis of the real symmetric matrix L_a. Retain the coordinates with eigenvalue $< c$, separately in each of the h real field coordinates. Complexification used in the Fourier proof preserves real eigenvalue multiplicities; it does not double them. Energy controls coefficients at eigenvalues $\ge c$, while $c\|C(U)\|^2$ controls the rest. Equation (8), $p = D/2$, and $(1+m)/m \le 2$ give (4). For $n = 1$ take the identity on all h coordinates. This constructs the actual global spectral observation. □

3 Every lower positive inverse exponent is insufficient

Theorem 2. *For every fixed $0 < r < D/2$, the analogue of Theorem 1 under $\operatorname{mean}((m/a)^r) \le 2$ is false uniformly in the actual scalar class $1 \le a \le 2m$, $\operatorname{mean}(a) = m$.*

Proof. Choose a fixed integer $\ell > 3D/c$, and put $P = 2\ell + 2$, $B = (2\ell+1)^D$. The triangular window $t(s) = (1 - |s|/\ell)_+$ has finite sums

$$T = \sum t^2 = (2\ell^2 + 1)/(3\ell), \quad V = \sum (d^- t)^2 = 2/\ell, \quad W = \sum (d^+ d^- t)^2 = 6/\ell^2.$$

For a product window $\psi_y(x) = \prod_i t(x_i - y_i)$ and nonzero actual adjoint H, the actual curl $U_1 = d_2^- \psi_y H$, $U_2 = -d_1^- \psi_y H$, other components zero, has backward divergence zero. Its field support and all nonzero forward-gradient start sites lie in the box $y + [-\ell, \ell]^D$. Direct separable sums give

$$\|U\|^2 = 2VT^{D-1}\|H\|^2, \qquad \frac{\mathcal{E}_0(U)}{\|U\|^2} = \frac{W}{V} + (D-1)\frac{V}{T} = \frac{3}{\ell} + \frac{6(D-1)}{2\ell^2+1} \le \frac{3D}{\ell} < c.$$

Take sufficiently large multiples n of P, set $m = n^2$ and $J = \lfloor n^{D-2r}/B \rfloor$. These J boxes fit on the separated P-grid eventually, since $P^D \le Bn^{2r}$. They satisfy $J \ge n^{D-2r}/(2B) \to \infty$. Put $s = BJ/n^D \le m^{-r}$ and set $a = 1$ on their union, $a = A = (m-s)/(1-s)$ elsewhere. Eventually $s \le 1/2$, so

$$\operatorname{mean}(a) = m, \quad 1 \le a \le 2m, \quad A \ge m, \qquad \operatorname{mean}((m/a)^r) \le sm^r + 1 \le 2.$$

Each curl's gradients start only in its weak box. The actual J-dimensional span has that same Rayleigh quotient, because both field and gradient supports are disjoint. Section 1 forces $\dim F \ge J$, whereas the target maximum at $m = n^2$ is 1. This proves the refutation using original nonlinear data. □

Together the two theorems give the sharp inverse-moment exponent for this specified family of global scalar conditions. They do not identify all sufficient coefficient criteria, prove a physical coefficient hypothesis, or impose spatial locality on C.

4 Spatial means and the critical dyadic obstruction

For a partition $\mathcal{P}$ into integer rectangles, its actual normalized means are

$$C_{\mathcal{P}}(U)_{R,j} = \sqrt{|R|}\,\mathrm{mean}_R\, U_j, \qquad \dim F = h\#\mathcal{P}.$$

A full dyadic cube tree splits each selected cube into all 2^D equal children and retains unsplit siblings. The origin of the torus coordinate cube may be translated. This class excludes arbitrary anisotropic rectangle cuts and compressed observations.

Proposition 1. *There are fixed critical-moment profiles with target maximum* 1 *for which every coercive full dyadic cube-tree mean partition has* $\Omega(\log n)$ *leaves, even after choosing a translated origin. The same profiles admit an explicit constant-size rectangular mean partition satisfying the global inequality for every actual cochain.*

Proof. Keep $\ell > 3D/c$ fixed. Choose a dyadic $L = 2^q \geq 4\ell + 2$, put $\gamma = \max(1, c/4)$, $n = 2^J$ with $J \geq q + 1$, $m = \gamma n^2$, and $S = \{0, \dots, L-1\}^D$. Set $a = 1$ on S and $a = A = (m - s)/(1 - s)$ elsewhere, $s = L^D/n^D$. Then $1 \leq a \leq 2m$, its mean is exactly m, and

$$\mathrm{mean}((m/a)^{D/2}) \leq 1 + L^D\gamma^{D/2} =: M_0.$$

All these parameters except n are fixed, and the target maximum is 1.

In the hypothetical complete level $j_* = J - q$ of any translated dyadic grid, cells have side L. Each coordinate interval of S meets at most two such cyclic grid intervals, so one intersection has integer length at least $L/2 \geq 2\ell + 1$. Taking these intersections independently places a whole gradient-start box for a compact curl inside S and inside one level-j_* cell. If the proposed partition had maximum depth at most j_*, that cell would lie in one actual leaf. The curl then has zero means on every leaf, since each component has total sum zero, yet its weighted Rayleigh quotient is $< c$. This contradicts coercivity. A single zero-extended window needs no packing divisibility assumption on n; $n \geq 2L$ suffices here. The field may depend on the partition, since coercivity is asserted for every field. Thus maximum depth is at least $J - q + 1$, and the full-arity tree identity gives

$$\#\mathcal{P} \geq 1 + (2^D - 1)(J - q + 1). \tag{9}$$

For a successful unrestricted partition, isolate all L^D vertices of S as singletons, and use the D disjoint complement rectangles

$$R_i = [0, L-1]^{i-1} \times [L, n-1] \times [0, n-1]^{D-i}.$$

Coverage follows from the unique first coordinate outside $[0, L-1]$. Every internal edge starts at a strong vertex with coefficient A. On a length-l interval, counting variance and path Cauchy–Schwarz give

$$\sum \|u - \bar{u}\|^2 \leq \frac{l(l-1)}{4A} \sum A\|du\|^2.$$

Indeed the path resistance is at most $(l-1)/A$, and a cut at t is crossed by $t(l-t) \leq l^2/4$ pairs in the variance identity. Product-variance tensorization on original fibres yields a rectangle constant $n^2/(4A)$, with no extra factor D. Singletons have zero residual, internal edge sets are disjoint, and omitted edges have positive energy. Since $cn^2/(4A) \leq c/(4\gamma) \leq 1$, exact orthogonal decomposition proves

$$c\|U\|^2 \leq Q_w(U) + c\|C_{\mathcal{P}}(U)\|^2, \qquad \#\mathcal{P} = L^D + D.$$

These rectangles have potentially unbounded aspect ratio; they leave the cube-tree class. □

The general critical moment question for unrestricted rectangular means remains open. Changing only the stopping rule within full cube trees cannot resolve it. Classical weighted Sobolev inequalities [2] also yield block observations for inverse moments of finite order strictly greater than $D/2$; the dossier contains the explicit balanced-box proof. Proposition 1 explains why its critical tree-counting loss cannot be removed merely by choosing a better cube-tree stopping criterion.

5 Evidence provenance and verification limits

The analytic sources, their exact revisions and the executable finite diagnostics are collected in a hash-allowlisted companion package. Its map gives full SHAs and SHA256 values; short forms below are identifiers, not substitute hashes.

Item	Source revision	Verification status
Actual flat Hodge identity	`dfd49d0b0730`	earlier formal source
Critical spectral theorem	`527fcdcd119a`	analytic adapters
Dyadic spatial separation	`fb65a67af360`	analytic adapters
Higher moment block proof	`6dcb893155e5`	analytic adapters
Exact finite fixture code	`4bb460e1537e`	own Colab109
Full diagnostic report	`602ae3f0b59a`	finite rational batch

Colab109 checked four actual profiles on $(\mathbb{Z}/2\mathbb{Z})^4$ using rational arithmetic. Each normal and optimized Python mode accepted four profiles and rejected 16 real matrix, inverse or flow mutations, with matching outputs and exit code zero. It checked the constant mode, forward symbols, actual pseudoinverse identities, feasible/minimizing flow energy, comparison inertia, strict low-spectrum multiplicities and noncommuting counts. There are 12 distinct density examples, evaluated again for each profile: 48 evaluations per mode, not 48 independent examples.

All eight files in the diagnostic archive's inner manifest were rehashed. Runtime execution was recorded at 17:13:06.300923–17:13:16.201862 UTC on 2 November2026, with automatic runtime release and zero sessions verified afterwards. Rounded resource units do not prove zero cost. No local Windows mathematical execution occurred. This single embedded-SHA finite batch is not two fresh matching clones, a Lean proof, a theorem about arbitrary dimensions or odd lattices, or independent mathematical review. No new Lean run occurs in this appendix's PDF assembly. Operational repository checks on unchanged Lean files do not verify these new analytic proofs.

6 Model assessments and review context

The prior analytic-source first self-assessments were 6/10 for the critical spectral note, 5/10 for the dyadic note, and 5/10 for the exact finite diagnostic package. They apply only to their own exact source versions. All were by the preparing Codex agent; the served backend identity and effort were unexposed. They are neither independent model reviews nor human peer review. No earlier score is inherited by this appendix. The first assessment of the exact compiled PDF is preserved separately with its SHA256, full rubric, objections and visual-review record. No grade-seeking repeated review, editorial acceptance or publication status is implied.

The principal limitations and objections are these. The new finite Fourier, inverse, field and partition adapters are analytic rather than kernel-verified. The classical Rumin–Frank method and elementary refinement geometry limit any novelty claim; a complete priority comparison is unfinished. The sufficient theorem uses one shared scalar coefficient profile, and its distribution

condition is not proved from arithmetic means. Spectral coordinates are global; unrestricted critical rectangular means remain unresolved. Exact finite diagnostics cover a small selected family, and some count bounds are loose. The constants are safe rather than sharp. This appendix makes no statement about an interacting or continuum physical mass gap. PR29 remains 20/41 and TermSource=0.

7 The single remaining bounded question

For fixed $D \geq 3, c > 0, N_c \geq 2, M$, do all actual scalar profiles $1 \leq a \leq 2m$, exact mean m, satisfying (3), admit (4) using normalized means on an unrestricted axis-aligned integer rectangular partition? The mechanism must leave full cube trees, for example through anisotropic cuts with a proved variance bound. A falsifier must defeat every such rectangular partition, not merely every dyadic cube tree. After three substantive units without a proved reduction or useful falsifier, the route must be changed or its concrete blocker identified. Formalization and consolidation of the existing results are preserved work, not additional active new research questions.

References

[1] R. L. Frank, *Cwikel's theorem and the CLR inequality*, arXiv:1206.3325v1 (2012), Section2, Theorem2.1, Corollary2.2, Lemmas2.4–2.5. `https://arxiv.org/abs/1206.3325`.

[2] S. Andres, J.-D. Deuschel and M. Slowik, *Harnack inequalities on weighted graphs and some applications to the random conductance model*, arXiv:1312.5473v6 (2019), Section2.2, equation(2.5) and Proposition2.1. `https://arxiv.org/abs/1312.5473`.